processFNIRS2: A Modular Open-Source MATLAB Toolbox for fNIRS Analysis, Validated on the FRESH Benchmark

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Synopsis

We present processFNIRS2 (pf2), an open-source MATLAB toolbox built on a modular, configuration-driven architecture spanning the full fNIRS workflow. A two-layer design separates single-subject processing from group analysis: a user-facing API package (`+pf2`) wraps an internal infrastructure layer (`+pf2_base`) and an interchangeable algorithm layer, while a dedicated group-analysis package (`+exploreFNIRS`) handles LME modeling, connectivity, and hyperscanning. Processing follows a three-stage pipeline (raw → optical density → hemoglobin → filtered) whose steps are composable value-class pipelines built from precomputed, immutable processing functions with dynamic argument injection. A device-abstraction layer supports 16 hardware configurations through declarative `.cfg` files, and a `ProcessingContext` object enables globals-free, reproducible, parallel execution. To confirm numerical correctness, we validated pf2 against three established toolboxes (Cedalion, MNE-NIRS, AnalyzIR) on the FRESH benchmark, obtaining activation-map correlations of r = 0.87–0.96.

Motivation

fNIRS analysis is fragmented across many toolboxes, each encoding different processing choices, data structures, and interfaces. The FRESH study (Yuecel et al., 2025) showed that 38 independent teams processing identical fNIRS datasets produced hypothesis pass rates ranging from 25% to 100%, underscoring how both implementation and analyst choices drive variability. A toolbox that makes processing decisions explicit, composable, and reproducible — while remaining interoperable with community standards — is therefore valuable for transparent, multi-site work. processFNIRS2 (pf2) is designed around these goals: a modular architecture in which every processing step is a first-class, inspectable object; a configuration-driven core that supports multiple devices without code changes; and an execution model that can bypass global state entirely for reproducible and parallel analyses. Here we describe the architecture and confirm, on the FRESH benchmark, that pf2 produces results numerically consistent with three independently developed platforms.

Architecture

Two-layer analysis model. pf2 is organized into two layers with a clean interface. Layer 1 (single-subject processing) imports raw device files and converts them to hemoglobin biomarkers; Layer 2 (`+exploreFNIRS`) consumes collections of processed structs for group statistics, connectivity, and hyperscanning. The contract between layers is a plain processed-fNIRS struct, so subjects processed by any means can enter group analysis.

Package layering. Within Layer 1, a user-facing API package (`+pf2`) provides import, export, data manipulation, probe/ROI, and plotting namespaces. It wraps an internal infrastructure layer (`+pf2_base`) — device loading, core fNIRS math, plotting style, and wavelet transforms — which in turn calls an interchangeable algorithm layer (`/functions`) of stand-alone processing kernels (motion correction, filtering, referencing). A configuration layer of declarative `.cfg` files sits beneath all of it. A progressive-disclosure API mirrors this hierarchy: parent-level calls (`pf2.import`) auto-select or prompt, while explicit child calls (`pf2.import.importSNIRF`) give direct control.

Three-stage pipeline. Processing is factored into three stages — raw intensity → optical density (motion correction, filtering, referencing), optical density → hemoglobin (modified Beer–Lambert with selectable DPF modes, including age-dependent DPF after Scholkmann and Wolf, 2013), and hemoglobin → filtered (baseline correction, filtering, artifact rejection, ROI averaging). Intermediate stages are retained, making each transformation inspectable and reproducible.

Composable pipeline value classes. Each processing step is a `PipelineFunction` — an immutable value object that precomputes its function handle and argument mapping. Ordered chains are `Pipeline` objects (with `RawPipeline`/`OxyPipeline` specializations for stages 1 and 3), supporting `add`, `insert`, `remove`, and `setParam`, and round-tripping to/from named method configs. At runtime a dynamic argument-injection mechanism supplies each kernel only the inputs it declares (data matrix, sampling rate, channel mask, markers, auxiliary signals, etc.), so generic functions compose without bespoke glue.

Device abstraction. A single codebase supports 16 fNIRS hardware configurations through INI-style `.cfg` files describing wavelengths, channel maps, and 2D/3D optode geometry. The `pf2.Device` value class exposes clean accessors (wavelengths, MNI positions, source–detector distances, channel tables) with a persistent cache, and is auto-attached to imported data.

Reproducible, globals-free execution. For interactive use, settings live in global state. For batch, parallel, and reproducible work, a `ProcessingContext` object encapsulates all settings, methods, and the device as locals; passing it to `processFNIRS2` bypasses globals entirely — they are neither read nor written — and the context serializes for exact reproduction. This enables isolated unit testing and one-context-per- worker parallelism.

Standards, visualization, and testing. pf2 imports and exports SNIRF and full BIDS-NIRS datasets, and exports a self-describing HDF5 tensor contract for foundation-model workflows. Theme-aware plotting renders topographic maps, 3D cortical surfaces, connectomes, and DOT reconstructions headlessly (saved figures always export on white). Correctness is guarded by a `matlab.unittest` suite — unit, integration, and quick-validation tests — backed by synthetic data generators for hemodynamics, noise, motion artifacts, and physiology.

Analysis Capabilities

On this architecture, pf2 provides seven motion correction methods (TDDR (Fishburn et al., 2019), wavelet (Molavi and Dumont, 2012), spline, SplineSG (Jahani et al., 2018), sSMART (Ayaz et al., 2010), spline+wavelet hybrid, and Takizawa rejection (Takizawa et al., 2008)); scalp-coupling-index quality control; short-channel regression; block averaging and GLM with OLS or AR-IRLS noise models; functional connectivity; hyperscanning with Hyper-Brain ICA (Luo et al., 2025); graph-theoretic network metrics; and diffuse optical tomography. Group analysis adds LME modeling with FDR correction and publication-ready visualization.

Validation

To confirm numerical correctness, we implemented matched pipelines across pf2, Cedalion (Python) (Middell et al., 2026), MNE-NIRS (Python) (Luke et al., 2021), and AnalyzIR (MATLAB) (Santosa et al., 2018) on both FRESH datasets (Yuecel et al., 2025): Dataset I (auditory speech-noise, $N = 17$, 33 channels) and Dataset II (motor finger-tapping, $N = 10$, 64 long-separation + 4 short-separation channels). All toolboxes used equivalent steps (SNIRF import, SCI threshold 0.75, motion correction, Beer–Lambert DPF = 6, bandpass 0.01–0.5 Hz, short-channel regression, block averaging, and AR-IRLS GLM). Twenty-two pf2 variants spanned five motion correction methods, two noise models, and with/without short-channel regression and SCI QC. Agreement used Pearson correlation of channel-wise activation, ICC(3,1) (Shrout and Fleiss, 1979), Sørensen–Dice similarity of significance maps, and Bland–Altman analysis.

Activation maps were highly correlated across all toolbox pairs (Figure **??**): $r = 0.87$–0.96 for auditory block averaging and 0.82–0.94 for standardized GLM, with Dice coefficients of 0.94–0.95 for motor significance maps. Raw ICC(3,1) was near zero (0.01–0.02) because DPF/PPF conventions produce up to 10^6-fold magnitude differences (μM vs. mol/L); after per-toolbox z-scoring, ICC rose to 0.86 (auditory) and 0.42 (motor). Individual-level motor pass rates were concordant across all four toolboxes (Table **??**): combined contralateral activation reached 80–100% (block averaging) and 70–100% (GLM), all at or above the FRESH median and within the 27th–100th percentile of the FRESH team distribution.

Table 1: Dataset II individual-level motor hypothesis pass rates (X/10 subjects). H1: left hand 2s FT activates contralateral PMC; H2: left + right 2s FT; H3: all sessions (2s+3s); H4: 3s > 2s duration effect. BA = block averaging, GLM = AR-IRLS.

Toolbox	Method	H1	H2	H3	H4
pf2	BA	4/10	9/10	9/10	1/10
	GLM	6/10	8/10	9/10	2/10
Cedalion	BA	5/10	10/10	10/10	1/10
	GLM	2/10	7/10	9/10	2/10
MNE-NIRS	BA	6/10	8/10	9/10	4/10
	GLM	5/10	7/10	9/10	7/10
AnalyzIR	BA	4/10	8/10	10/10	6/10
	GLM	8/10	10/10	10/10	0/10
FRESH median		5/10	8/10	9/10	2/10

Discussion

The architecture makes processing decisions explicit and composable: pipelines are inspectable objects, devices are declarative configs, and a `ProcessingContext` guarantees that a serialized analysis reproduces byte-for-byte without global-state coupling. The FRESH validation confirms this design is not only transparent but numerically faithful — pf2 implements core fNIRS algorithms (Beer–Lambert conversion, TDDR, bandpass filtering, GLM estimation) consistently with three independently developed platforms. The highest correlation was between pf2 and AnalyzIR ($r = 0.96$), both MATLAB toolboxes sharing numerical libraries, yet pf2 also correlated strongly with the Python toolboxes ($r = 0.87$–0.92), indicating the agreement is not an artifact of shared implementation. The near-zero raw ICC highlights that DPF/PPF and unit conventions — not algorithmic disagreement — dominate apparent cross-toolbox differences; we recommend documenting these conventions explicitly and using scale-invariant metrics for cross-platform comparison. The concordant motor results reinforce the FRESH conclusion that analytical choices drive more variability than toolbox implementation. We will extend this comparison to connectivity and hyperscanning metrics at the conference.

comparison/output/datasetI_correlation.png

Figure 1: Inter-toolbox Pearson correlation of channel-wise group-level HbO activation (Dataset I, block averaging). All pairwise correlations exceed r = 0.75, with pf2 correlations ranging from 0.87 (MNE-NIRS) to 0.96 (AnalyzIR).

Availability

processFNIRS2 is open-source (GPLv3) and released with documentation, runnable tutorials, and the comparison harness used for this validation.

Disclosures

H. Ayaz was involved in the technology development of fNIR Devices, LLC and was thus offered a minor share in the startup firm. fNIR Devices, LLC manufactures optical brain imaging instruments and licensed IP and know-how from Drexel University.

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comparison/output/datasetII_brain3d_topfront.png

Figure 2: Group-level motor activation (t-statistic) from GLM AR-IRLS with short-channel regression (Dataset II), rendered with pf2's built-in headless 3D cortical visualization. Bilateral motor cortex activation is consistent with the finger-tapping paradigm. Sources (yellow) and detectors (red) shown.

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